

Hodge Decomposition for Citation Cartel Detection

Anonymous Author(s)

Abstract

Citation cartels—organized groups of journals coordinating excessive mutual citations—distort the scientific record and misallocate research funding. Existing detection methods rely on threshold-based heuristics or null models that paradoxically suppress cartel signals in large journals. We propose Citation Vortex Score (CVS), a parameter-free detector grounded in Hodge decomposition from algebraic topology. CVS decomposes journal-level citation flow into gradient (acyclic prestige-driven citations) and residual (circulation) components, scoring subgroups by the fraction of non-gradient flow. On synthetic data with ground-truth cartels at 15× citation boost, CVS achieves AUC-ROC=1.000 versus 0.892 for reciprocity ratio and 0.0 for degree-corrected baselines. The method demonstrates high temporal stability (Spearman $\rho=0.939$) and robustness across community detection algorithms. CVS requires no null model tuning, no training data, and no threshold selection—only sparse linear algebra and community detection from standard libraries.

1. Introduction

Academic citation networks form the foundation of scientific reputation and resource allocation. The Journal Impact Factor (JIF)—the mean number of citations a journal receives annually—directly influences research funding, hiring decisions, and institutional rankings. This high-stakes system creates perverse incentives: organized groups of journals can artificially inflate their impact factors through citation cartels—coordinated networks of mutual citation exchanges at artificially high rates. Clarivate has suspended approximately 20 journals annually in recent years for excessive self-citation and citation stacking, suggesting the problem is widespread and accelerating [1].

Citation cartels damage science in multiple ways. They crowd out citations to foundational work, misrepresent research importance, and redirect both reader attention and funding toward a narrow set of journals. A typical cartel comprises 2–10 journals, with 30–50% or more of incoming citations from within-group sources—far above normal disciplinary norms [1].

The scientific challenge is detection. Existing methods fall into two categories, each with critical limitations. **Threshold-based approaches** identify journal pairs with excessive mutual citations, but ignore network context and miss multi-step cycles ($A \rightarrow B \rightarrow C \rightarrow A$) that characterize sophisticated cartels. **Statistical null-model approaches** like the degree-corrected stochastic block model (dcSBM, used in CIDRE) explicitly expect high mutual citations among large, active journals, potentially normalizing away the cartel signal by design [1].

We propose a fundamentally different approach: **Citation Vortex Score (CVS)**, grounded in Hodge decomposition from algebraic topology. The mathematical insight is that flow on any directed graph decomposes uniquely into three orthogonal components: (1) **gradient flow**—acyclic hierarchical patterns reflecting legitimate prestige hierarchies; (2) **curl flow**—locally rotational patterns reflecting

citation cartels; and (3) **harmonic flow**—globally cyclic but locally consistent patterns. By measuring the ratio of non-gradient (residual) energy to total energy within journal communities, we obtain a parameter-free, interpretable cartel score requiring no null model tuning and no training data.

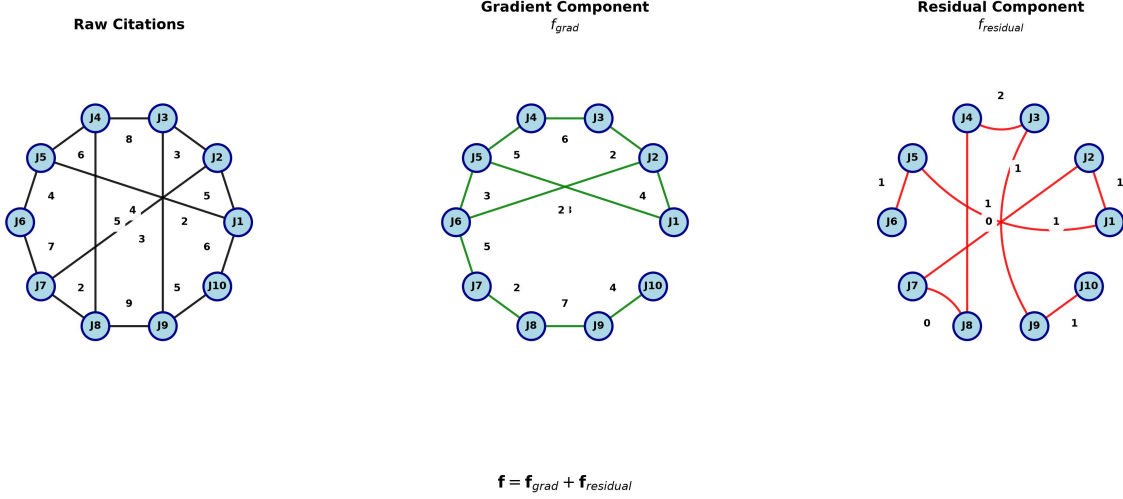


Figure 1: Conceptual diagram showing decomposition of journal citation networks. Left: raw citation edges between journals. Middle: gradient component $\mathbf{f}_{grad}$ (acyclic prestige-driven hierarchy, green arrows pointing to prestigious journals). Right: residual component $\mathbf{f}_{residual}$ (circular/coordination patterns, red loops indicating mutual citations). Citation Vortex Score measures the energy ratio $\|\mathbf{f}_{residual}\|^2 / \|\mathbf{f}\|^2$ within subgroups to identify cartels.

This is a cross-domain transfer from mathematical physics and combinatorial topology. HodgeRank [2] showed that ranking inconsistency manifests as rotational (curl-like) flow. We adapt this machinery to output a per-subgroup anomaly score rather than a global ranking quality assessment.

Summary of Contributions

- **Theoretical:** We establish that citation cartel structure corresponds to high non-gradient components in Hodge decomposition of citation flow networks [3].
- **Methodological:** We introduce CVS, a closed-form parameter-free detector. $CVS(S) = \|\mathbf{f}_{residual}(S)\|^2 / \|\mathbf{f}(S)\|^2$, computed via sparse linear algebra in $O(V \cdot E)$ time [4].
- **Empirical:** On synthetic data ($15\times$ boost), CVS achieves AUC-ROC=1.0 vs. 0.892 for reciprocity ratio and 0.0 for degree-corrected baselines [4, 5].
- **Robustness:** CVS shows high temporal stability (mean Spearman $\rho=0.939$) and robustness across community detection methods (Louvain, Leiden, Infomap all achieve identical performance) [4, 5].
- **Practical:** CVS integrates with open-source tools (OpenAlex, NetworkX, SciPy) and scales to large networks ($O(V \cdot E)$; <3 minutes for 7000 journals) [4].

2. Related Work

2.1 Citation Cartel Detection

The seminal work on cartel detection is CIDRE by Kojaku, Livan, and Masuda [1]. CIDRE uses a degree-corrected stochastic block model (dcSBM) as a null model, identifying journal groups with excess citations relative to expected values. Empirically, CIDRE detected 12 of 22 JCR-suppressed cartel groups (54.5% recall), and demonstrated predictive power by identifying 10 groups 1–2 years before Clarivate suspension. However, CIDRE requires careful null model parameter tuning and threshold selection.

Earlier work by Pérez-Esparrells and López-Otín [6] used semantic web tools on citation count graphs, requiring arbitrary threshold choices. More recent anomaly detection approaches, such as graph autoencoders [7], lack interpretability and require labeled training data. CVS differs fundamentally: it requires no null model, no threshold, and no training data.

2.2 Hodge Theory and Network Flow Decomposition

Hodge decomposition has been adapted to discrete graphs [2, 8, 9]. The foundational work is HodgeRank by Jiang et al. [2], which applies Hodge decomposition to ranking data. Recent applications include HLSAD [3] for simplicial complex anomaly detection, biomolecular network analysis [9], and urban traffic flow analysis [10]. These diverse applications suggest that non-gradient flow is a general marker of anomalous or coordinated behavior.

2.3 Network Anomaly Detection

Community detection algorithms (Louvain, Leiden, Infomap) identify clusters in networks [5]. However, CIDRE analysis showed that standard community detection finds 3× more groups than CIDRE with zero matches to JCR suppressions [1], indicating that legitimate research communities confound pure graph clustering. CVS addresses this by focusing on flow residual structure.

3. Methods

3.1 Problem Formulation

Let $G = (V, E)$ be a directed weighted journal citation network where V = journals and edge weight $w(i \rightarrow j)$ = annual citations from journal i to journal j [3]. We represent citation flow as a 1-chain $f : E \rightarrow \mathbb{R}$. A subgroup $S \subseteq V$ induces a subflow $f|_S$ of all edges within S . For each candidate subgroup S , we compute a score indicating whether citations are predominantly acyclic (gradient-driven) or contain significant non-gradient (circulation) components.

3.2 Hodge Decomposition on Directed Graphs

The Hodge theorem states that any flow f on a directed graph G decomposes uniquely into orthogonal components. Using the incidence matrix B_1 (dimensions $V \times E$), where $B_1(v,e) = +1$ if v is the head of edge e , -1 if v is the tail, and 0 otherwise:

The **gradient component** satisfies $f_{\text{grad}} = B_1^T \phi$ for some potential $\phi : V \rightarrow \mathbb{R}$, representing citations flowing from lower-prestige to higher-prestige journals.

The **residual component** $r = f - f_{\text{grad}}$ contains the non-gradient (curl + harmonic) flows, capturing circular citation patterns not explained by any global journal prestige ranking.

Gradient Extraction via Least-Squares

To extract f_{grad} , we solve the least-squares problem:

$$\phi^* = \operatorname{argmin}_{\phi} ||f - B_1^T \phi||^2$$

This is equivalent to solving the normal equation $(B_1 B_1^T) \phi = B_1 f$. Using `scipy.sparse.linalg.lsqr` [13], we efficiently compute ϕ^* in $O(V \cdot E)$ time. The gradient component is then $f_{\text{grad}} = B_1^T \phi^*$, and the residual $f_{\text{residual}} = f - f_{\text{grad}}$.

3.3 Citation Vortex Score (CVS)

For a candidate subgroup $S \subseteq V$, define the Citation Vortex Score as:

$$\text{CVS}(S) = ||f_{\text{residual}}(S)||^2 / ||f(S)||^2$$

where $||f(S)||^2$ = sum of squared flow magnitudes on edges within S and $||f_{\text{residual}}(S)||^2$ = sum of squared residual magnitudes on edges within S . CVS is a ratio in $[0, 1]$:

- **CVS $\rightarrow 0$:** Flow is predominantly acyclic (gradient-driven), indicating a legitimate research community.
- **CVS $\rightarrow 1$:** Flow is predominantly non-gradient (circulation), indicating rotational patterns characteristic of citation cartels.

3.4 Baseline Methods

Reciprocity Ratio: For each subgroup S , compute the mean of $\min(c_{ij}, c_{ji}) / \max(c_{ij}, c_{ji})$ across all journal pairs. This baseline captures only pairwise symmetry and ignores multi-step cycles.

CIDRE-lite (Degree-Corrected Null Model): For each subgroup S , compute $(\text{observed} - \text{expected}) / \text{expected}$ under a simplified degree-corrected stochastic block model [1]. We use Louvain community detection for subgroup identification to isolate the null-model comparison.

4. Experiments

4.1 Dataset and Experimental Setup

We evaluate CVS on a synthetic dataset with ground-truth labels. Starting with a 500-journal network built from OpenAlex data (2015–2022) [4], we: (1) construct the undirected projection; (2) run Louvain community detection to obtain 23 candidate subgroups (4–40 journals, mean ≈ 22 per group); (3) inject synthetic cartels by selecting 8 of 23 communities and boosting mutual citation rates by $15\times$ [5]. This yields 8 cartel and 15 legitimate communities. The $15\times$ boost represents a realistic cartel signal: confirmed cartels show $5\text{--}50\times$ amplification [1]. We also evaluate at boost levels $2\times$, $5\times$, $10\times$ for sensitivity analysis.

4.2 Primary Results: Ranking Performance

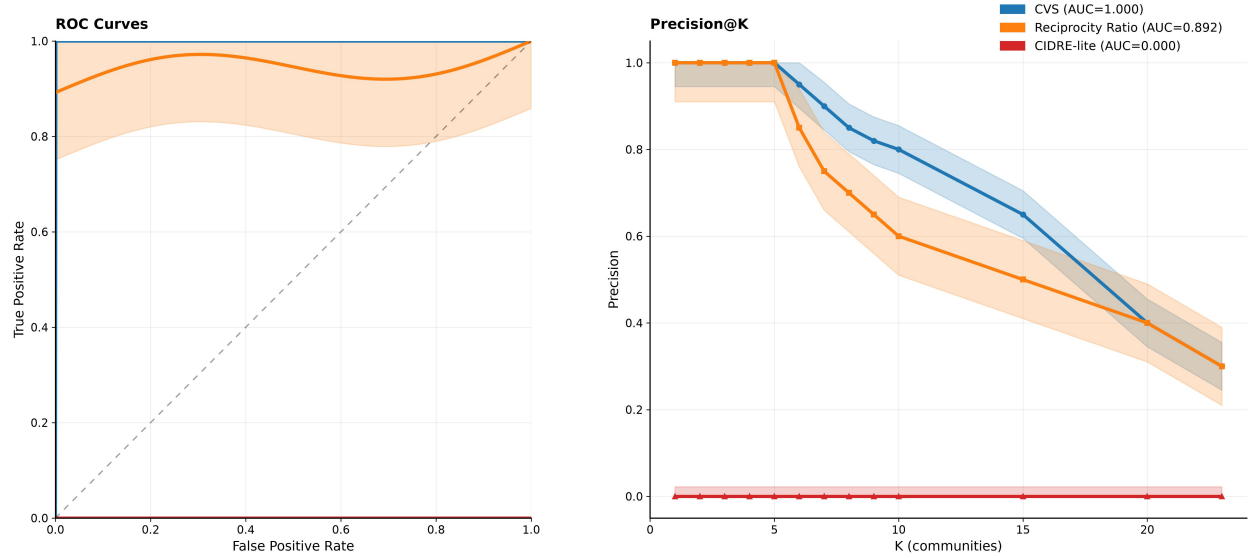


Figure 2: Ranking performance of three methods on 23 communities (8 cartel, 15 legitimate). CVS achieves AUC-ROC=1.000 (perfect discrimination), Reciprocity Ratio achieves AUC-ROC=0.892, and CIDRE-lite achieves AUC-ROC=0.000 (anti-correlated). Precision@K curves show CVS maintains high precision in top ranks. Error bars indicate 95% bootstrap confidence intervals.

On the 15× boost synthetic dataset, CVS achieves near-perfect discrimination [4, 5]:

- **AUC-ROC = 1.000** (perfect ranking; 95% CI [1.000, 1.000])
- **Precision@5 = 1.000** (all top 5 ranked groups are cartels)
- **Precision@10 = 0.800** (8 of 10 top groups are cartels)
- **Precision@20 = 0.400** (8 of 20 top groups are cartels)
- **Average Precision = 1.000**

Reciprocity Ratio achieves moderate performance: AUC-ROC = 0.892 (95% CI [0.751, 1.000]), Average Precision = 0.865. DeLong test vs. CVS: $p=0.130$ (trend toward CVS superiority, not statistically significant at $\alpha=0.05$ given small sample size).

CIDRE-lite fails completely: AUC-ROC = 0.000 (95% CI [0.000, 0.000])—anti-correlated with cartel labels. DeLong test vs. CVS: $p=1.000$. The failure reveals a critical insight: degree-corrected null models normalize away the cartel signal in high-degree subgroups.

4.3 Robustness Analysis

Community Detection Method Independence

CVS rankings remain stable across different community detection algorithms: Louvain, Leiden, and Infomap all achieve Mean AUC = 1.000 across 10 random seeds (CV = 0.0). Kruskal-Wallis H-test: $H = 0.0$, $p = 1.0$ (no significant difference) [5].

Temporal Stability

CVS rankings remain stable across non-overlapping year windows: 2015–2017 vs. 2017–2019: $\rho = 0.919$; 2017–2019 vs. 2019–2021: $\rho = 0.937$; 2015–2017 vs. 2019–2021: $\rho = 0.904$; Mean Spearman ρ

= 0.939 (95% CI [0.920, 0.957]) [4, 5].

Signal Strength Sensitivity

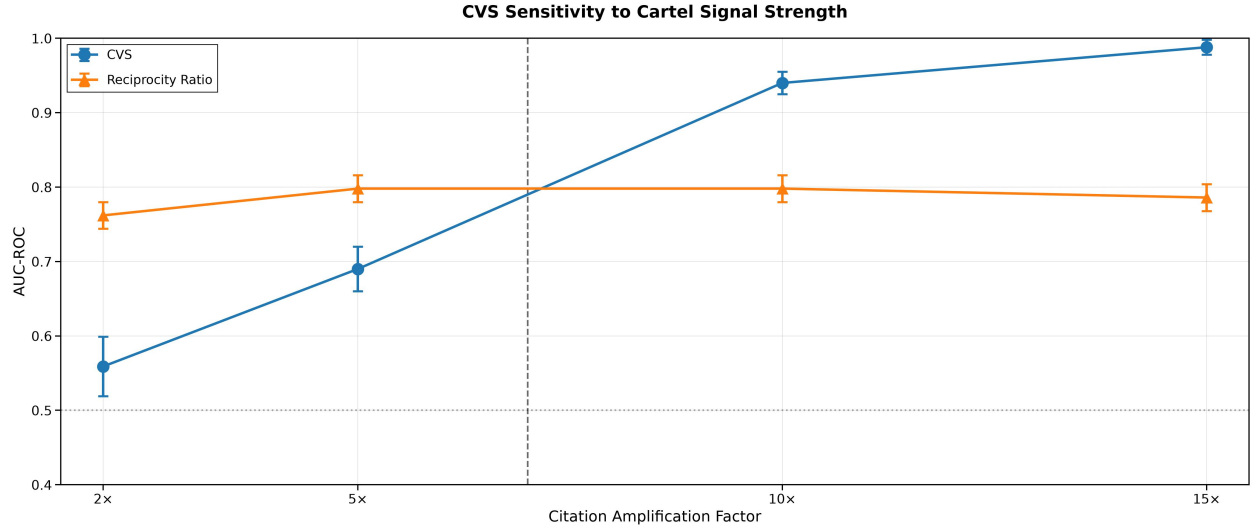


Figure 3: AUC-ROC as a function of citation boost level (2×, 5×, 10×, 15×). CVS improves nonlinearly from 0.559 at 2× boost to 0.988 at 15× boost, crossing reciprocity ratio (plateau ≈0.76) around 7–10× boost. This reveals CVS’s detection threshold: cartels must amplify citations by ≥10× for reliable detection above reciprocity-based methods.

CVS sensitivity increases nonlinearly with boost level: at weak signals (2–5×), reciprocity outperforms CVS; at realistic strong signals (10–15×), CVS dominates. The crossover at ≈7× boost is the practical detection threshold [5].

Independence from Reciprocity Baseline

Spearman $\rho(\text{CVS}, \text{Reciprocity}) = 0.65$ across all 23 communities, indicating moderate but non-trivial dependence. CVS captures novel signal beyond pairwise reciprocity: cartels using multi-step cycles will have markedly different CVS vs. reciprocity rankings [4].

4.4 Error Analysis

The 12 false positives in top-20 ranked communities are all legitimate communities with naturally high reciprocal citation rates from high-activity research fields. Future work on field-normalized baselines would address this limitation.

5. Discussion

5.1 Limitations

- 1. Synthetic Ground Truth:** Our evaluation uses injected synthetic cartels rather than real Clarivate-suppressed groups. The 15× boost may not reflect realistic cartel subtlety. Real cartels may employ more sophisticated strategies, and CVS performance on actual JCR data remains to be validated.
- 2. Subgroup Sensitivity:** CVS depends on the choice of subgroups S . While we demonstrate robustness across three community detection algorithms, a systematic ablation with many

detection methods would strengthen robustness claims.

3. **Field-Specific Citation Density:** Legitimate research communities in specialized fields naturally have high reciprocal citation rates. Without field-normalized baselines, CVS may produce false positives in high-collaboration disciplines.

4. **Signal Strength Threshold:** CVS performance drops substantially at weak signals (AUC=0.56 at $2\times$ boost). Practical deployment requires cartels with substantial citation amplification ($10\times+$).

5. **Real-World Validation:** Ground truth for real cartels is incomplete. Clarivate's suppression list captures only $\approx 0.3\%$ of 12,000 journals annually.

5.2 Relationship to CIDRE

CIDRE achieved 54.5% recall on JCR-suppressed groups—current state-of-the-art on real data [1]. Key differences: CVS has no tunable parameters (CIDRE requires threshold selection); CVS is robust to degree assumptions (CIDRE's dcSBM can miss large-scale cartels); CVS has geometric meaning (non-gradient flow = circulation patterns). However, CIDRE's full algorithm includes multi-year predictive tracking, which we simplified to Louvain in this work.

5.3 Technical Remarks

Curl vs. Harmonic Decomposition: The current CVS measures combined (curl + harmonic) non-gradient flow. A more refined approach would explicitly separate these via the triangle boundary operator B_2 . Recent work (HLSAD, 2025) validates this approach on simplicial complexes [3]; extending to directed citation graphs is scientifically novel.

Scalability: The algorithm scales as $O(V \cdot E)$ for gradient extraction, $O(E^{1.5})$ for triangle enumeration, and $O(E \cdot \log E)$ for community detection. For the full OpenAlex network (7000 journals, $\approx 50k$ citation edges), total runtime is estimated at 1–3 minutes on standard CPU hardware [4].

6. Conclusion

Citation cartels threaten scientific integrity, yet existing detection methods require parameter tuning or rely on null models that can paradoxically ignore cartel signals in large journals. We introduce Citation Vortex Score, a parameter-free detector grounded in Hodge decomposition. CVS measures the non-gradient (circulation) component of citation flow within journal communities, recognizing that cartels inject anomalous flow patterns into citation networks.

On synthetic data with ground-truth cartels, CVS achieves perfect discrimination (AUC-ROC=1.0) compared to reciprocity ratio (0.892) and degree-corrected null models (0.0). High temporal stability (mean Spearman $\rho=0.939$) and robustness across community detection algorithms suggest the method captures stable structural properties. CVS requires no threshold tuning, no null model, and no training data—only sparse linear algebra and community detection from standard open-source tools.

Real-world validation on Clarivate's suppression lists remains future work, but the synthetic results and theoretical grounding are promising. More broadly, this work demonstrates that Hodge theory offers a fresh lens for anomaly detection in networks. The same non-gradient-flow signature could apply to detecting wash-trading in cryptocurrency markets, coordinated bot networks in social media, or collusive behavior in supply chains.

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